Chapter 5: Finite and Infinite Horizon Models

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5.1.1 The setup

- The individual chooses a consumption path $\{c(\tau)\}$ over $[t, T]$.
- Wealth is the state variable; consumption is the control variable.

$$U(t) = \int_t^T e^{-\rho[\tau - t]} \ln c(\tau)\, d\tau, \tag{5.1.1}$$

$$\dot{a}(\tau) = r(\tau)a(\tau) + w(\tau) - c(\tau). \tag{5.1.2}$$

5.1.2 Solving by optimal control

- Use the current-value Hamiltonian.
- The costate variable $\lambda(\tau)$ corresponds to the state variable $a(\tau)$.

$$H = \ln c(\tau) + \lambda(\tau)\big[r(\tau)a(\tau) + w(\tau) - c(\tau)\big]. \tag{5.1.3}$$

$$\frac{\partial H}{\partial c} = \frac{1}{c} - \lambda = 0, \tag{5.1.4}$$

$$\dot{\lambda} = \rho\lambda - \frac{\partial H}{\partial a} = \rho\lambda - r\lambda. \tag{5.1.5}$$

5.1.2 Solving by optimal control: Euler equation

- From the first-order condition, $\lambda = 1/c$.
- Taking logs and differentiating gives the Keynes-Ramsey rule.

$$\ln \lambda = -\ln c,$$
$$\frac{\dot{\lambda}}{\lambda} = -\frac{\dot{c}}{c}.$$

$$-\frac{\dot{c}}{c} = \rho - r \quad \Longleftrightarrow \quad \frac{\dot{c}}{c} = r - \rho. \tag{5.1.6}$$

- Equations (5.1.2) and (5.1.6) form a two-dimensional differential equation system in (a, c).

5.2.1 The setup

- The Hamiltonian is derived as a shortcut from the Lagrangian approach.
- The control path $\{z(\tau)\}$ is chosen subject to a law of motion for $y(\tau)$.

$$U(t) = \int_t^T e^{-\rho[\tau - t]} u(y(\tau), z(\tau), \tau)\, d\tau, \tag{5.2.1}$$

$$\dot{y}(\tau) = Q(y(\tau), z(\tau), \tau). \tag{5.2.2}$$

5.2.2 Solving by the Lagrangian

- There is one constraint for each instant τ.
- The multiplier $\eta(\tau)$ is time-dependent.

$$L = \int_t^T e^{-\rho[\tau - t]} u(\tau)\, d\tau + \int_t^T \eta(\tau)\big[Q(\tau) - \dot{y}(\tau)\big]\, d\tau.$$

$$L = \int_t^T \Big(e^{-\rho[\tau - t]} u(\tau) + \eta(\tau) Q(\tau)\Big)\, d\tau + \int_t^T \dot{\eta}(\tau) y(\tau)\, d\tau - [\eta(\tau) y(\tau)]_t^T. \tag{5.2.3}$$

5.2.2 Solving by the Lagrangian: Control

- Maximize the transformed Lagrangian with respect to $z(\tau)$ at each instant.
- Define the current-value costate variable $\lambda(\tau)$.

$$e^{-\rho[\tau-t]}u_z(\tau) + \eta(\tau)Q_z(\tau) = 0,$$
$$u_z(\tau) + e^{\rho[\tau-t]}\eta(\tau)Q_z(\tau) = 0.$$

$$\lambda(\tau) \equiv e^{\rho[\tau-t]}\eta(\tau), \tag{5.2.4}$$
$$u_z(\tau) + \lambda(\tau)Q_z(\tau) = 0. \tag{5.2.5}$$

5.2.2 Solving by the Lagrangian: State

- Maximize with respect to the state variable $y(\tau)$.
- Transform the resulting condition into a law of motion for $\lambda(\tau)$.

$$\begin{aligned} e^{-\rho[\tau-t]}u_y(\tau) + \eta(\tau)Q_y(\tau) + \dot{\eta}(\tau) &= 0, \\ \dot{\lambda}(\tau) &= \rho e^{\rho[\tau-t]}\eta(\tau) + e^{\rho[\tau-t]}\dot{\eta}(\tau). \end{aligned} \tag{5.2.6}$$

$$\dot{\lambda}(\tau) = \rho\lambda(\tau) - u_y(\tau) - \lambda(\tau)Q_y(\tau). \tag{5.2.7}$$

5.2.3 Hamiltonians as a shortcut

- The current-value Hamiltonian adds instantaneous payoff and the costate-weighted law of motion.
- The shortcut reproduces (5.2.5) and (5.2.7).

$$H = u(\tau) + \lambda(\tau)Q(\tau). \tag{5.2.8}$$

$$H_z = 0, \tag{5.2.9}$$

$$\dot{\lambda} = \rho\lambda - H_y. \tag{5.2.10}$$

5.2.3 Hamiltonians as a shortcut: Shadow price

- Define the value function as the maximized objective from initial state $y(t)$.
- The costate variable is the shadow value of the state variable.

$$V(y(t)) \equiv \max_{\{z(\tau)\}} U(t),$$
$$V'(y(t)) = \lambda(t).$$

- As in static Lagrange problems, λ measures the marginal value of relaxing the state/endowment.

5.3.1 Solving by optimal control

- Replace the finite terminal date T by an infinite horizon.
- The objective must converge along all feasible paths.

$$\begin{aligned} &\max_{\{z(\tau)\}} \int_t^{\infty} e^{-\rho[\tau-t]} u(y(\tau), z(\tau), \tau)\, d\tau, \\ &\text{s.t.} \quad \dot{y}(\tau) = Q(y(\tau), z(\tau), \tau), \qquad y(t) = y_t. \end{aligned} \tag{5.3.1}$$

- With $H = u(\tau) + \lambda(\tau)Q(\tau)$, the necessary conditions remain

$$\frac{\partial H}{\partial z} = 0, \qquad \dot{\lambda} = \rho\lambda - \frac{\partial H}{\partial y}.$$

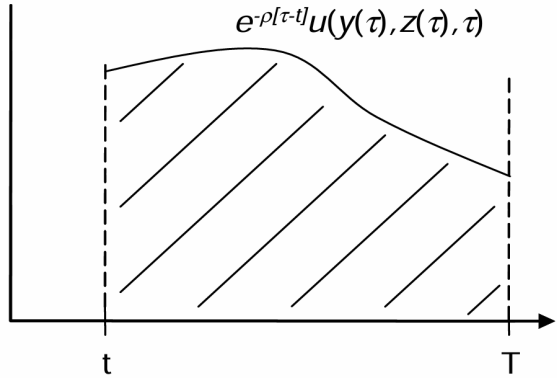


Figure 5.3.1 *Bounded objective function for a finite horizon*

5.3.2 The boundedness condition

- Infinite-horizon utility must be finite; otherwise the optimization criterion is ill-defined.
- Use CES instantaneous utility and assume consumption grows at rate g.

$$u(c(\tau)) = \frac{c(\tau)^{1-\sigma} - 1}{1-\sigma}, \qquad \sigma > 0, \tag{5.3.2}$$
$$c(\tau) = c(t)e^{g[\tau - t]}.$$

$$U(t) = \frac{1}{1-\sigma}\left[c(t)^{1-\sigma}\int_t^{\infty} e^{-\rho[\tau-t]}e^{(1-\sigma)g[\tau-t]}\,d\tau + \frac{1}{\rho}\right].$$

5.3.2 The boundedness condition: Result

- The integral converges only if discounted instantaneous utility falls sufficiently fast.

$$\int_t^{\infty} e^{[(1-\sigma)g-\rho][\tau-t]}\, d\tau < \infty \quad \Longleftrightarrow \quad (1-\sigma)g - \rho < 0.$$

Boundedness condition

$$(1-\sigma)g < \rho.$$

- Instantaneous utility grows at rate $(1-\sigma)g$; impatience discounts at rate ρ.

5.4 Boundary conditions and sufficient conditions

- Without boundary conditions, the differential equation system admits infinitely many paths.
- Boundary conditions pin down levels of the state and control variables.

$$\text{Example system:} \qquad \dot{a} = ra + w - c, \qquad \frac{\dot{c}}{c} = r - \rho.$$

- The chapter considers free endpoint, fixed endpoint, and infinite-horizon conditions.

5.4.1 Free value of the state variable at the endpoint

- The initial state is given; the terminal state is freely chosen.
- The terminal condition is imposed on the costate variable.

$$y(t) = y_t, \quad y(T) \text{ free.} \tag{5.4.1}$$

$$\lambda(T) = 0. \tag{5.4.2}$$

- Together with $y(t) = y_t$, this provides the two boundary conditions for the system.

5.4.2 Fixed value of the state variable at the endpoint

- Both initial and terminal states are imposed directly.
- Feasibility of the terminal condition must be checked.

$$y(t) = y_t, \qquad y(T) = y_T. \tag{5.4.3}$$

- Difference from the free-endpoint case: $y(T) = y_T$ is exogenous.
- In the free-endpoint case, $\lambda(T) = 0$ is endogenous as a necessary optimality condition.

5.4.3 The transversality condition

- Infinite-horizon problems lack a finite terminal date T.
- A practical boundary condition can come from the no-Ponzi-game restriction.

$$\lim_{T\to\infty} y(T)\exp\left[-\int_t^T r(\tau)\,d\tau\right] = 0, \qquad r \equiv \frac{\partial H}{\partial y}.$$

$$\lim_{T\to\infty} e^{-\rho T}\lambda(T)y(T) = \lim_{T\to\infty} \eta(T)y(T) = 0. \tag{5.4.4}$$

5.4.3 The transversality condition: Abstract form

- Another route is to require convergence to a well-defined steady state.
- An abstract TVC is often written in terms of the present-value costate.

$$\lim_{t\to\infty} \left\{\eta(t)\left[y(t) - y^*(t)\right]\right\} = 0,$$

- $y^*(t)$ denotes the optimal path of the state variable.
- In applications, saddle-path convergence is often used to select the equilibrium path.

5.4.4 Sufficient conditions

- Conditions derived so far are necessary; sufficiency requires curvature restrictions.
- For problems (5.2.1)–(5.2.2), the necessary conditions are sufficient under any of:

1. $u(\cdot)$ and $Q(\cdot)$ are concave in (y, z) and $\eta(\tau) > 0$ for all τ.
2. $Q(\cdot)$ is linear in (y, z) for any $\eta(\tau)$.
3. $Q(\cdot)$ is convex and $\eta(\tau) < 0$ for all τ.

- In many economic applications, the shadow price is positive because utility increases in consumption.

5.5.1 A firm with adjustment costs

- Capital is not rented instantaneously; installation is costly.
- Investment affects future capital, making the firm's problem intertemporal.

$$\Pi_0 = \int_0^T e^{-rt}\pi(t)\,dt, \tag{5.5.1}$$

$$\dot{K}(t) = I(t) - \delta K(t), \tag{5.5.2}$$

$$\pi(t) = pF(K(t)) - \Phi(I(t)). \tag{5.5.3}$$

- Assumptions: $F'(\cdot) > 0$, $F''(\cdot) < 0$, and $K(0) = K_0$.

5.5.1 A firm with adjustment costs: Hamiltonian

- The current-value Hamiltonian combines profits and capital accumulation.

$$
\begin{aligned}
H &= \pi(t) + \lambda(t)[I(t) - \delta K(t)] \\
&= pF(K(t)) - \Phi(I(t)) + \lambda(t)[I(t) - \delta K(t)].
\end{aligned}
$$

$$
H_I = -\Phi'(I(t)) + \lambda(t) = 0, \tag{5.5.4}
$$

$$
\dot{\lambda}(t) = r\lambda - H_K(t) = (r + \delta)\lambda - pF'(K(t)). \tag{5.5.5}
$$

5.5.1 A firm with adjustment costs: Investment dynamics

- Use the first-order condition to eliminate the costate variable.

$$\lambda(t) = \Phi'(I(t)),$$
$$\dot{\lambda}(t) = \Phi''(I(t))\dot{I}(t).$$

$$\Phi''(I(t))\dot{I}(t) = (r+\delta)\Phi'(I(t)) - pF'(K(t)). \tag{5.5.6}$$

- Equations (5.5.2) and (5.5.6) form a two-dimensional system in (K, I).

5.5.1 A firm with adjustment costs: Example

- Let adjustment costs be linear-quadratic in investment.

$$\Phi(I) = v\left(I + \frac{I^2}{2}\right). \tag{5.5.7}$$

$$\Phi'(I) = v(1 + I), \qquad \Phi''(I) = v.$$

$$\dot{I}(t) = (r + \delta)(1 + I(t)) - \frac{p}{v}F'(K(t)). \tag{5.5.8}$$

- Together with $\dot{K} = I - \delta K$, this gives the phase-diagram system.

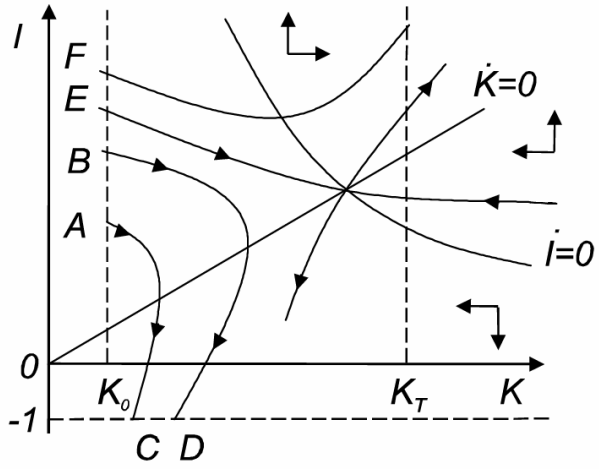


Figure 5.5.1 *A firm with adjustment cost*

5.5.2 Free value at the end point

- If the terminal capital stock is free, impose $\lambda(T) = 0$.
- In the adjustment-cost example, this pins down terminal investment.

$$\lambda(T) = 0, \qquad \lambda(T) = \Phi'(I(T)).$$

$$\Phi'(I(T)) = 0 \quad \Longleftrightarrow \quad v[1 + I(T)] = 0 \quad \Longleftrightarrow \quad I(T) = -1.$$

- The optimal trajectory is the one that reaches $I(T) = -1$ exactly at the terminal date.

5.5.3 Fixed value at the end point

- Suppose the endpoint requires a fixed capital stock K_T.
- Feasibility must be checked: some K_T may require too much investment.

$$K(0) = K_0, \qquad K(T) = K_T.$$

- From $\dot{K} = I - \delta K$, high enough investment can mechanically raise K.
- But profits $\pi(t) = pF(K(t)) - \Phi(I(t))$ may become negative at very high $I(t)$.

5.5.4 Infinite horizon and transversality condition

- With $T = \infty$, the model needs a long-run boundary condition.
- A natural requirement is convergence to a steady state of firm size.

$$\begin{aligned} \dot{K} = 0 \quad &\Longleftrightarrow \quad I = \delta K, \\ \dot{I} = 0 \quad &\Longleftrightarrow \quad (r + \delta)(1 + I) = \frac{p}{v} F'(K). \end{aligned}$$

- The optimal path starts on the saddle path leading asymptotically to the steady state.
- If a shock moves the firm off the saddle path, investment jumps back to the saddle path.

5.6.1 Infinite horizon – optimal consumption paths

- Consider household utility over an infinite horizon.
- A budget constraint is needed; otherwise optimal consumption would be infinite.

$$U(t) = \int_t^\infty e^{-\rho[\tau - t]} u(c(\tau))\, d\tau. \tag{5.6.1}$$

$$\int_t^\infty D_r(\tau) E(\tau)\, d\tau = a(t) + \int_t^\infty D_r(\tau) w(\tau)\, d\tau, \tag{5.6.2}$$

$$E(\tau) = p(\tau) c(\tau), \qquad D_r(\tau) = \exp\left[- \int_t^\tau r(u)\, du \right].$$

5.6.1 Infinite horizon – solving by Lagrangian

- The intertemporal budget constraint is a single constraint.
- Therefore the Lagrange multiplier λ is time-independent.

$$L = \int_t^\infty e^{-\rho[\tau - t]} u(c(\tau))\, d\tau - \lambda \left[\int_t^\infty D_r(\tau) E(\tau)\, d\tau - a(t) - \int_t^\infty D_r(\tau) w(\tau)\, d\tau \right].$$

$$L_{c(\tau)} = e^{-\rho[\tau - t]} u'(c(\tau)) - \lambda D_r(\tau) p(\tau) = 0. \tag{5.6.3}$$

5.6.1 Infinite horizon – Keynes-Ramsey rule

- Take logs of (5.6.3) and differentiate with respect to τ.

$$-\int_t^\tau r(u)\,du - \ln p(\tau) + \rho[\tau - t] = \ln \lambda - \ln u'(c(\tau)).$$

$$-\frac{u''(c(\tau))}{u'(c(\tau))}\dot{c}(\tau) = r(\tau) - \frac{\dot{p}(\tau)}{p(\tau)} - \rho. \tag{5.6.4}$$

- The curvature term governs the intertemporal elasticity of substitution.

5.6.1 Infinite horizon – logarithmic utility

- For $u(c(\tau)) = \ln c(\tau)$, $u'(c) = 1/c$ and $u''(c) = -1/c^2$.

$$\frac{\dot{c}(\tau)}{c(\tau)} = r(\tau) - \frac{\dot{p}(\tau)}{p(\tau)} - \rho. \tag{5.6.5}$$

- Consumption grows when the real return exceeds the time-preference rate.

5.6.1 Infinite horizon – Hamiltonian formulation

- The same problem can be written with a dynamic budget constraint.
- This formulation is suited to the Hamiltonian method.

$$\dot{a}(t) = r(t)a(t) + w(t) - p(t)c(t). \tag{5.6.6}$$

$$H = u(c(t)) + \lambda(t)\big[r(t)a(t) + w(t) - p(t)c(t)\big].$$

- The resulting Euler equation is equivalent to (5.6.4).

5.6.1 The interest rate effect on consumption

- Assume CRRA utility, constant r, and $p = 1$.
- The intertemporal budget constraint and Keynes-Ramsey rule become:

$$\int_t^\infty e^{-r[\tau - t]} c(\tau)\, d\tau = a(t) + \int_t^\infty e^{-r[\tau - t]} w(\tau)\, d\tau, \tag{5.6.7}$$

$$\frac{\dot{c}(\tau)}{c(\tau)} = \frac{r - \rho}{\sigma}. \tag{5.6.8}$$

5.6.1 The interest rate effect: Solving consumption growth

- Solve the linear differential equation in consumption.

$$c(\tau) = c(t)e^{\frac{r-\rho}{\sigma}(\tau - t)}. \tag{5.6.9}$$

$$\begin{aligned}\int_t^\infty e^{-r[\tau - t]}c(\tau)\,d\tau &= c(t)\int_t^\infty e^{-\left(r - \frac{r-\rho}{\sigma}\right)[\tau - t]}\,d\tau \\ &= \frac{\sigma c(t)}{\rho - (1-\sigma)r}.\end{aligned}$$

- This requires $r > (r-\rho)/\sigma$, equivalently $(1-\sigma)r < \rho$.

5.6.1 The interest rate effect: Consumption level

- Insert the previous expression into the intertemporal budget constraint.

$$c(t) = \frac{\rho - (1-\sigma)r}{\sigma} \left\{ a(t) + \int_t^{\infty} e^{-r[\tau - t]} w(\tau)\, d\tau \right\}. \tag{5.6.10}$$

- Consumption is a fraction of total wealth: financial wealth plus human wealth.
- For $\sigma = 1$, the fraction equals ρ.

$$\frac{\partial c(t)}{\partial r} = -\frac{1-\sigma}{\sigma}\{\cdot\} \quad \gtrless \quad 0 \quad \Longleftrightarrow \quad \sigma \gtrless 1.$$

5.6.1 The boundary condition for the infinite horizon

- The dynamic budget constraint and Keynes-Ramsey rule give two differential equations.
- The initial wealth stock gives one boundary condition.
- The no-Ponzi-game condition gives the second boundary condition.

$$\dot{a} = ra + w - pc,$$
$$\frac{\dot{c}}{c} = r - \frac{\dot{p}}{p} - \rho \quad \text{or} \quad \frac{r-\rho}{\sigma}.$$

- The no-Ponzi condition yields the intertemporal budget constraint and hence the initial level $c(t)$.

5.6.2 Necessary conditions, solutions and state variables

- A necessary condition gives information about changes; a solution gives levels.
- Compare the Keynes-Ramsey rule with the closed-form consumption rule.

Necessary condition:
$$\frac{\dot{c}(\tau)}{c(\tau)} = \frac{r-\rho}{\sigma}. \tag{5.6.8}$$

Solution:
$$c(t) = \frac{\rho-(1-\sigma)r}{\sigma}\left\{a(t) + \int_t^{\infty} e^{-r[\tau-t]} w(\tau)\, d\tau\right\}. \tag{5.6.10}$$

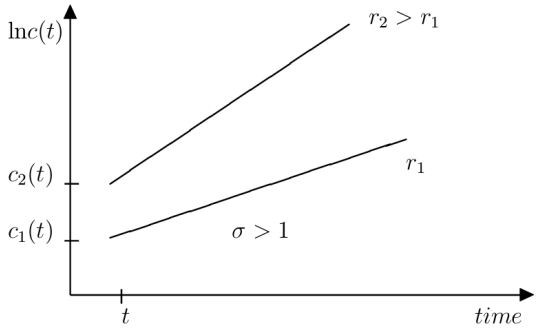


Figure 5.6.1 *The effect of the interest rate on consumption growth and consumption level for an intertemporal elasticity of substitution smaller than one, i.e.* $\sigma > 1$

5.6.2 Necessary conditions, solutions and state variables: Closed loop

- A closed-loop solution expresses the control as a function of states and time.
- An open-loop solution expresses the control as a function of time only.

$$c(t) = c\left(a(t), \int_t^{\infty} e^{-r[\tau - t]} w(\tau)\, d\tau\right).$$

- In (5.6.10), $a(t)$ and the future wage path affect current consumption.
- In deterministic settings, the wage path can be reduced to an initial value plus a time function.

5.6.3 Optimal growth – the central planner

- The planner chooses aggregate consumption to maximize social welfare.
- The felicity function has constant intertemporal elasticity of substitution $1/\sigma$.

$$\max_{\{C(\tau)\}} \int_t^\infty e^{-\rho[\tau-t]} u(C(\tau))\, d\tau,$$

$$u(C) = \frac{C^{1-\sigma} - 1}{1-\sigma}. \tag{5.6.11}$$

$$\dot{K}(t) = Y(K(t), L) - \delta K(t) - C(t). \tag{5.6.12}$$

5.6.3 Optimal growth – Keynes-Ramsey rule

- The current-value Hamiltonian is

$$H = u(C(t)) + \lambda(t)[Y(K(t), L) - \delta K(t) - C(t)].$$

$$\begin{aligned} u'(C(t)) &= \lambda(t), \\ \dot{\lambda}(t) &= \rho\lambda(t) - \lambda(t)[Y_K(K(t), L) - \delta]. \end{aligned} \tag{5.6.13}$$

$$-\frac{u''(C(t))}{u'(C(t))}\dot{C}(t) = Y_K(K(t), L) - \delta - \rho.$$

5.6.3 Optimal growth – Classic result

- With CES utility, $-u''(C)/u'(C) = \sigma/C$.

$$\frac{\dot{C}(t)}{C(t)} = \frac{Y_K(K(t), L) - \delta - \rho}{\sigma}. \tag{5.6.14}$$

- Consumption grows if marginal productivity exceeds depreciation plus impatience.
- Higher intertemporal elasticity $1/\sigma$ raises the response of consumption growth.

5.6.3 Optimal growth – Phase diagram

- The resource constraint and Keynes-Ramsey rule define a two-dimensional system.
- Zero-motion lines are:

$$\dot{K}(t) \geq 0 \quad \Longleftrightarrow \quad C(t) \leq Y(K(t), L) - \delta K(t), \tag{5.6.15}$$

$$\dot{C}(t) \geq 0 \quad \Longleftrightarrow \quad Y_K(K(t), L) \geq \delta + \rho. \tag{5.6.16}$$

- The initial capital stock is fixed; initial consumption jumps to the saddle path.

5.6.3 Optimal growth – Boundary selection

- The state variable $K(t)$ is historically given and cannot jump.
- The control variable $C(t)$ can jump to select the optimal path.

$$K(t) = K_0, \qquad C(t) = C_0 \text{ chosen to satisfy TVC/saddle-path convergence.}$$

- Paths above or below the saddle path violate the transversality condition or a no-Ponzi condition.
- In practice, convergence to the steady state is used to select the saddle path.

5.6.4 The matching approach to unemployment

- Unemployment changes through job destruction and job creation.
- Matches depend on unemployed workers U and vacancies V.

$$\dot{U} = sL - m(U, V). \tag{5.6.17}$$

$$L = N - U, \qquad \dot{U} = s[N - U] - m(U, V).$$

- With unemployment rate $u \equiv U/N$, constant returns imply

$$\dot{u} = s[1 - u] - m(u, V/N).$$

5.6.4 The matching approach: Tightness

- Define labour-market tightness by $\theta \equiv V/U$.
- Let $q(\theta)$ be the rate at which vacancies are filled.

$$m\left(u, \frac{V}{N}\right) = m\left(1, \frac{V}{U}\right) u = \frac{V}{U} m\left(\frac{U}{V}, 1\right) u \equiv \theta q(\theta) u. \tag{5.6.18}$$

$$\dot{u} = s[1 - u] - \theta q(\theta) u.$$

$$\dot{L} = m(N - L, V) - sL. \tag{5.6.19}$$

5.6.4 The matching approach: Firms

- A firm chooses vacancies rather than employment directly.
- This parallels adjustment costs: employment changes only gradually through matches.

$$\max_{\{V_i, K_i\}} \int_t^\infty e^{-r[\tau - t]} \pi_i(\tau)\, d\tau,$$
$$\pi_i = Y(K_i, L_i) - rK_i - wL_i - \kappa V_i.$$

$$\dot{L}_i = q(\theta) V_i - sL_i. \tag{5.6.20}$$

5.6.4 The matching approach: Firm FOCs

- The current-value Hamiltonian is $H = \pi_i + \lambda_i \dot{L}_i$.

$$H = Y(K_i, L_i) - rK_i - wL_i - \kappa V_i + \lambda_i[q(\theta)V_i - sL_i].$$

$$H_{K_i} = Y_{K_i}(K_i, L_i) - r = 0, \qquad H_{V_i} = -\kappa + \lambda_i q(\theta) = 0. \tag{5.6.21}$$

$$\dot{\lambda}_i = (r + s)\lambda_i - Y_{L_i}(K_i, L_i) + w. \tag{5.6.22}$$

5.6.4 The matching approach: General equilibrium

- From the vacancy FOC, $\kappa = q(\theta)\lambda$.
- Differentiating gives $\dot{\lambda}/\lambda = -\dot{q}(\theta)/q(\theta)$.

$$-\frac{\dot{q}(\theta)}{q(\theta)} = r + s - \frac{q(\theta)}{\kappa}\big[Y_L(K, L) - w\big]. \tag{5.6.23}$$

- Equation (5.6.23), together with (5.6.19), determines vacancies and employment given boundary conditions.

5.6.4 The matching approach: Cobb-Douglas matching

- Assume constant returns Cobb-Douglas matching.

$$m = U^{\alpha} V^{\beta}, \qquad q(\theta) = \frac{m(N-L, V)}{V} = (N-L)^{\alpha} V^{\beta-1}.$$

$$\frac{\dot{q}(\theta)}{q(\theta)} = \frac{\dot{m}}{m} - \frac{\dot{V}}{V} = \alpha \frac{\dot{U}}{U} - (1-\beta)\frac{\dot{V}}{V}.$$

$$\frac{\dot{U}}{U} = s\frac{L}{N-L} - \frac{V^{\beta}}{(N-L)^{1-\alpha}}.$$

5.6.4 The matching approach: Vacancy dynamics

- Insert the Cobb-Douglas expression into the vacancy equation.

$$(1-\beta)\frac{\dot{V}}{V} = r + s - \frac{(N-L)^{\alpha}}{\kappa V^{1-\beta}}[Y_L(K,L) - w] + \alpha\left[s\frac{L}{N-L} - \frac{V^{\beta}}{(N-L)^{1-\alpha}}\right].$$

- This equation and $\dot{L} = m(N-L,V) - sL$ form a two-dimensional system in (V,L).

5.7.1 Problems without discounting

- Present-value Hamiltonians apply directly to objectives without explicit current-value transformation.
- The state and control vectors are $y(\tau) \in \mathbb{R}^n$ and $z(\tau) \in \mathbb{R}^m$.

$$\max_{z(\tau)} \int_t^T F(y(\tau), z(\tau), \tau)\, d\tau, \tag{5.7.1}$$

$$\dot{y}(\tau) = Q(y(\tau), z(\tau), \tau), \tag{5.7.2}$$

$$y(t) = y_t, \qquad y(T) \text{ free.} \tag{5.7.3}$$

5.7.1 Problems without discounting: PV Hamiltonian

- Define the present-value Hamiltonian and the vector of costate variables.

$$H^P = F(\tau) + \eta(\tau)Q(\tau), \tag{5.7.4}$$
$$\eta(\tau) = (\eta_1(\tau), \eta_2(\tau), \ldots, \eta_n(\tau)). \tag{5.7.5}$$

$$H_z^P = 0, \tag{5.7.6}$$
$$\dot{\eta}(\tau) = -H_y^P, \tag{5.7.7}$$
$$\eta(T) = 0.$$

5.7.1 Problems without discounting: Structure

- The m equations in (5.7.6) solve controls as functions of states and costates.

$$z(\tau) = z(y(\tau), \eta(\tau)).$$

- The $2n$ equations in (5.7.7) and (5.7.2) form a differential equation system.
- The n initial state conditions plus n terminal costate conditions solve the system.

5.7.2 Deriving laws of motion

- Add a static constraint $G(y(\tau), z(\tau), \tau) = 0$.

$$G(y(\tau), z(\tau), \tau) = 0. \tag{5.7.8}$$

$$L = \int_t^T \Big[F(:) + \eta(\tau)(Q(:) - \dot{y}(\tau)) - \xi(\tau) G(:) \Big] \, d\tau.$$

$$L = \int_t^T \Big[F(:) + \eta(\tau) Q(:) + \dot{\eta}(\tau) y(\tau) - \xi(\tau) G(:) \Big] \, d\tau - [\eta(\tau) y(\tau)]_t^T.$$

5.7.2 Deriving laws of motion: FOCs

- Maximize the transformed Lagrangian with respect to $z(\tau)$ and $y(\tau)$.

$$F_z(:) + \eta(\tau)Q_z(:) - \xi(\tau)G_z(:) = 0, \tag{5.7.9}$$
$$F_y(:) + \eta(\tau)Q_y(:) + \dot{\eta}(\tau) - \xi(\tau)G_y(:) = 0. \tag{5.7.10}$$

$$\dot{\eta}(\tau) = -F_y(:) - \eta(\tau)Q_y(:) + \xi(\tau)G_y(:). \tag{5.7.11}$$

5.7.2 Deriving laws of motion: Shortcut

- Define the present-value Hamiltonian with the static constraint included.

$$H^P = F(:) + \eta(\tau)Q(:) - \xi(\tau)G(:). \tag{5.7.12}$$

$$H_z^P = 0, \tag{5.7.13}$$

$$\dot{\eta}(\tau) = -H_y^P. \tag{5.7.14}$$

- These reproduce (5.7.9) and (5.7.11).

5.7.3 The link between CV and PV

- Solve the discounted problem using a present-value Hamiltonian.

$$H^P(\tau) = e^{-\rho[\tau - t]} G(:) + \eta(\tau) Q(:). \tag{5.7.15}$$

$$\eta(T) = 0, \tag{5.7.16}$$

$$\frac{\partial H^P}{\partial z} = e^{-\rho[\tau - t]} G_z(:) + \eta Q_z(:) = 0, \tag{5.7.17}$$

$$\dot{\eta}(\tau) = -\frac{\partial H^P}{\partial y} = -e^{-\rho[\tau - t]} G_y(:) - \eta Q_y(:). \tag{5.7.18}$$

5.7.3 The link between CV and PV: Simplification

- Define the current-value costate variable.

$$G_z(\tau) + \lambda(\tau) Q_z(\tau) = 0, \tag{5.7.19}$$

$$\lambda(\tau) \equiv e^{\rho[\tau - t]} \eta(\tau). \tag{5.7.20}$$

$$e^{\rho[\tau - t]} \dot{\eta}(\tau) = \dot{\lambda}(\tau) - \rho \lambda(\tau).$$

$$\dot{\lambda}(\tau) = \rho \lambda(\tau) - G_y(:) - \lambda(\tau) Q_y(:). \tag{5.7.21}$$

5.7.3 The link between CV and PV: Hamiltonians

- Multiplying the present-value Hamiltonian by $e^{\rho[\tau-t]}$ yields the current-value Hamiltonian.

$$\begin{aligned} e^{\rho[\tau-t]}H^P(\tau) &= G(:) + \lambda(\tau)Q(:) \\ &\equiv H^c(\tau). \end{aligned}$$

Link between Hamiltonians

$$H^c(\tau) = e^{\rho[\tau-t]}H^P(\tau).$$

- The current-value and present-value first-order conditions are equivalent.